Lab Exercise 19- ClusterIP in Service in Kubernetes

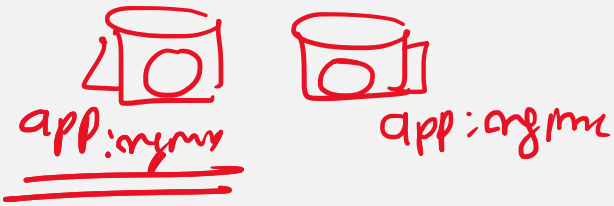
A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
      app: nginx  # MUST match service selector
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```



Save file.

```
kubectl apply -f nginx-deployment.yaml
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ notepad nginx-deployment.yaml

namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-app created

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Verify:

```
kubectl get pods --show-labels
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$  kubectl get pods --show-labels
NAME                          READY   STATUS    RESTARTS   AGE   LABELS
configmap-pod                 1/1     Running   0          21h   <none>
nginx-app-86c57bc6b8-gtv67    1/1     Running   0          24s   app=nginx,pod-template-hash=86c57bc6b8
nginx-app-86c57bc6b8-ld7pk    1/1     Running   0          24s   app=nginx,pod-template-hash=86c57bc6b8
test-vol1                     1/1     Running   0          20m   <none>
test-vol2                     1/1     Running   0          12m   <none>

namit@ThinkPadE15 MINGW64 ~ (main)
$ |
```

You should see:

```
app=nginx
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

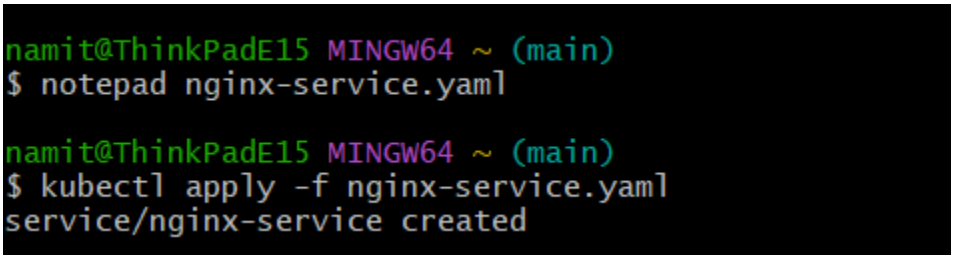
nginx-service.yaml

```
apiVersion: v1
kind: Service
metadata:
```

```
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
  - port: 80
    targetPort: 80
```

Apply:

```
kubectl apply -f nginx-service.yaml
```



Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

You should see:

```
TYPE        CLUSTER-IP
ClusterIP   10.x.x.x
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl get svc
NAME            TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)   AGE
kubernetes      ClusterIP   10.96.0.1     <none>        443/TCP   35d
nginx-service   ClusterIP   10.109.2.39   <none>        80/TCP    10s

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

Now you should see something like:

```
10.244.x.x:80, 10.244.x.x:80
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl get endpoints nginx-service
NAME            ENDPOINTS                     AGE
nginx-service   10.1.0.111:80,10.1.0.112:80   2m17s

namit@ThinkPadE15 MINGW64 ~ (main)
$
```

Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

Inside pod:

```
wget -qO- http://nginx-service
```

You should see:

```
Welcome to nginx!
```

Exit:

```
exit
```

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ kubectl run test-pod --image=busybox --rm -it -- sh
If you don't see a command prompt, try pressing enter.
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit
Session ended, resume using 'kubectl attach test-pod -c test-pod -i -t' command when the pod is running
pod "test-pod" deleted

namit@ThinkPadE15 MINGW64 ~ (main)
$ |
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.

```
namit@ThinkPadE15 MINGW64 ~ (main)
$ curl http://   10.109.2.39
curl: (3) URL rejected: No host part in the URL
|
```